



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab experiment-16

Experiment 16. Prolog Program to Sum Integers from 1 to N

Aim

To write a Prolog program to find the sum of integers from 1 to N.

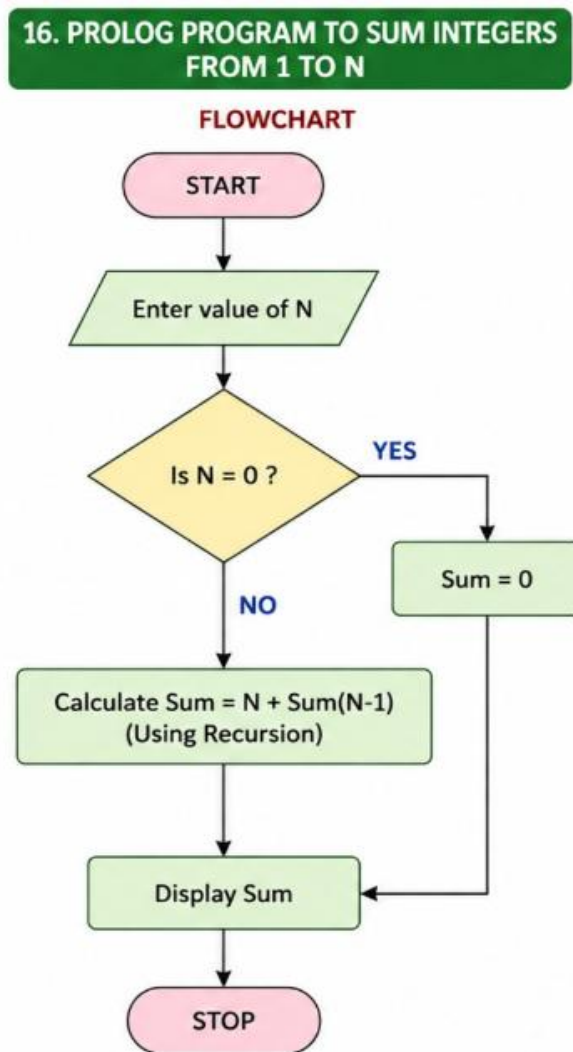
Objectives

- To understand recursion in Prolog.
- To calculate the sum of numbers.
- To use Prolog rules and facts.

Algorithm

1. Read the value of N.
2. If N is 0, return 0.
3. Otherwise calculate $N + \text{Sum}(N-1)$.
4. Continue until N becomes 0.
5. Display the sum.

Flowchart



Code

`sum(0, 0).`

`sum(N, S) :-`

`N > 0,`

N1 is N - 1,

sum(N1, S1),

S is N + S1.

`:- initialization(main).`

`main :-`

`sum(10, S),`

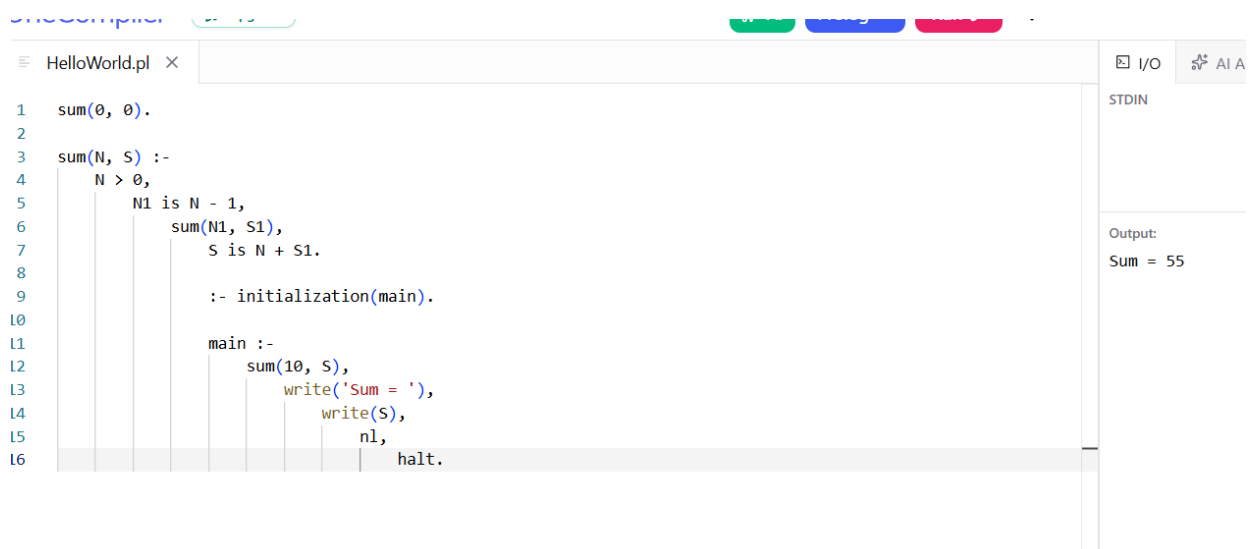
`write('Sum = '),`

`write(S),`

`nl,`

`halt.`

Output



```
1  sum(0, 0).
2
3  sum(N, S) :-
4      N > 0,
5      N1 is N - 1,
6      sum(N1, S1),
7      S is N + S1.
8
9      :- initialization(main).
10
11
12  main :-
13      sum(10, S),
14      write('Sum = '),
15      write(S),
16      nl,
17      halt.
```

STDIN

Output:
Sum = 55

Result

The Prolog program successfully calculates the sum of integers from **1 to N**.

Conclusion

Thus, the sum of integers from 1 to N was successfully implemented using recursion in Prolog.